User Guide | COMFORT ARF8275ARA | LoRaWAN EU863-870

Document version	V1.0
Product/Service	COMFORT
Area	LoRaWAN EU863-870
Zone	

PRODUCTS AND REGULATORY INFORMATION

💡 This User Guide applies to the following product:

COMFORT ARF8275AR LoRaWAN EU863-870
Reference:
ARF8275ARA

Firmware version: 1.2.X

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INTRODUCTION

https://s3-us-west-2.amazonaws.com/secure.notion-static.com/8d868d75-933f-409e-8902-4ddf2dc87f30/EU_declaration_of_conformity_(_Comfort_LoRaWAN_ARF8275AR).pdf

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1. PRODUCT PRESENTATION

1.1. General Description

The COMFORT by Adeunis is a connected device for measuring **temperature** and **ambient humidity** in a building.

This ready-to-use, sleek, and compact device measures thermal comfort data and sends it once via a **LPWAN network**. The measurement and data transmission frequencies are configurable to adapt to various deployment contexts.

It also features an alert button and a dry contact input to detect one or multiple events and send an associated frame.

The COMFORT is powered by a replaceable battery pack and can operate for more than **15 years without maintenance**.

This product is compatible with Adeunis' **Device Management** platform KARE and the KARE+ service for Over-The-Air updates of a sensor fleet. This adeunis sensor management offering allows for **operational cost optimization** by timely on-site interventions and avoiding unnecessary travels, **strengthening a business model** by ensuring the products' lifespan and adjusting their configuration, and **enhancing customer satisfaction** by enabling uninterrupted service.

1.2. Features

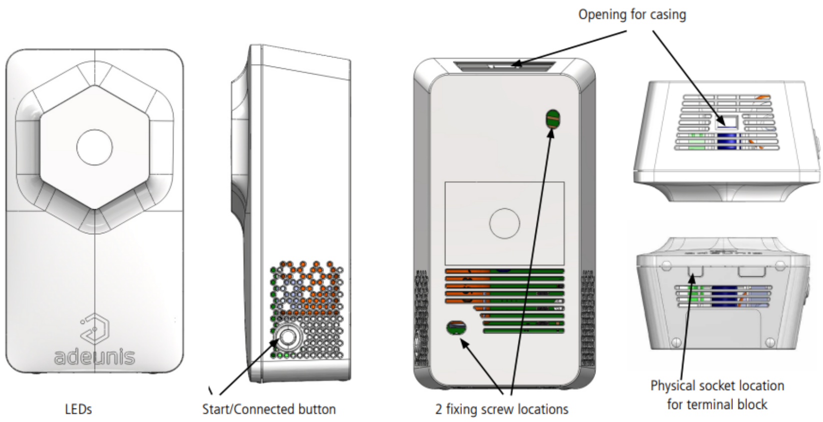
- **2-in-1 Sensor:** Temperature, humidity level
- **Dry Contact Input:** Allows for connecting an additional sensor to complement its functions (e.g., door opening detection, water detection)
- **Alert Button:** Detects one or multiple events and sends an associated frame (by button press)
- **Data Transmission Modes:** Periodic and/or event-based (exceeding high and/or low thresholds)
- **Battery-powered Standalone Product:** Powered by an internal replaceable battery pack designed to function for multiple years without replacement (refer to the autonomy table).
- **Autonomy Optimization:** Data logging (up to 16 samples per frame)

- **Local and Remote Configuration:** The product can be configured by the user locally via a USB-C port or remotely via the LoRaWAN network, allowing for parameter setting of periodicity, transmission modes, and alarm thresholds.

1.3. Package Contents

The product is delivered in a cardboard packaging containing the following elements: Front panel, back panel, electronic board, and ER18505H battery pack.

1.4. Casing



IMPORTANT NOTE 1

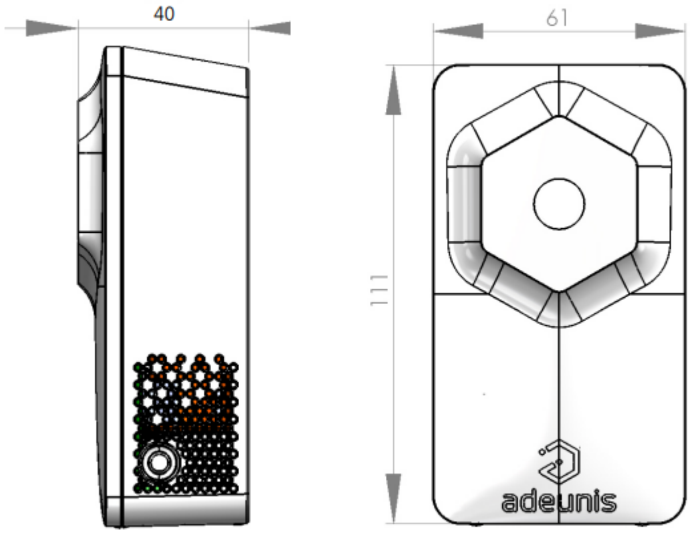
The COMFORT is delivered by default with an OTAA configuration, allowing the user to register their product with a LoRaWAN operator.

IMPORTANT NOTE 2

The COMFORT can be powered on using the button on the side of the housing or the IoT Configurator.

1.5 Dimensions

Values are in millimeters.



1.6. Push Button

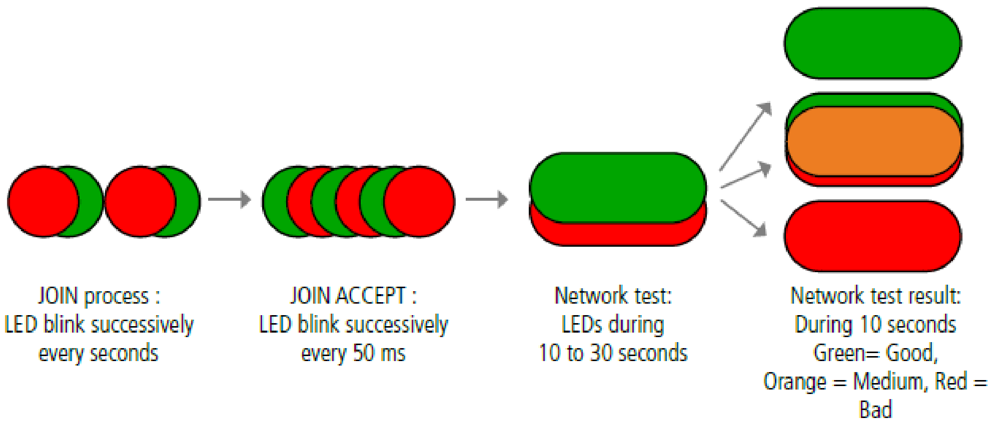
Function	Action
Product Startup	Long button press (> 5 seconds) when the product is in PARK mode (i.e., turned off)
Alarm Trigger	Short button press (< 500 milliseconds) when the product is in PRODUCTION mode (i.e., started)

1.7. LED Indicators

Mode	Red LED State	Orange LED State	Green LED State
Product in PARK mode	OFF	OFF	OFF
Long button press (> 5 sec) in PARK mode	OFF	OFF	ON for 1 second after button press
Product Startup	OFF	OFF	Blinking: 5 cycles of 100 ms ON / 100 ms OFF
Transition to COMMAND mode	OFF	ON	OFF
LoRaWAN JOIN Process	Blinking: 50 ms ON / 1 sec OFF	OFF	Blinking: 50 ms ON / 1 sec OFF (after red LED)

Mode	Red LED State	Orange LED State	Green LED State
JOIN Process: JOIN ACCEPT	Blinking: 50 ms ON / 50 ms OFF (6x)	OFF	Blinking: 50 ms ON / 50 ms OFF (6x) (before red LED)
Radio Quality Test - In progress	OFF	ON for 10 to 30 seconds	OFF
Radio Quality Test - Result	Poor Test: ON for 10 seconds	Average Test: ON for 10 seconds	Good Test: ON for 10 seconds
Event Detection: Short button press (< 500 ms) in PRODUCTION mode	OFF	ON during button press	OFF
Product Fault (factory return)	ON	OFF	OFF

LEDs scenario for a sensor configured in Class A OTAA:



1.8. Power Supply

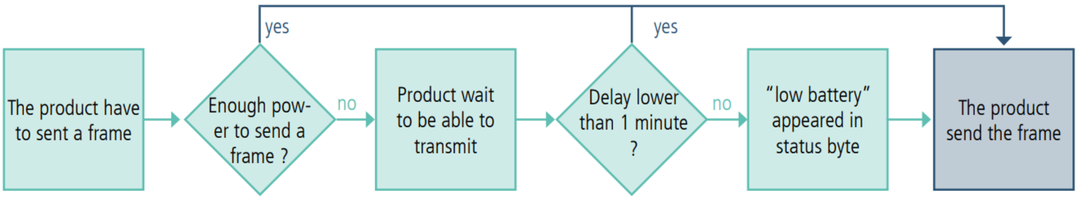
1.8.1 Power supply type

The COMFORT must be powered by a battery-pack FANSO Li-SOCI2 ER18505H.

- The USB-C port cannot be used to recharge the battery

1.8.2 Low battery management

When the product detects that the battery is unable to deliver the required power for transmission (extreme temperatures or end of battery life), it waits until it is capable of transmitting. If it detects that the resulting delay is longer than 1 minute, it notifies the user through the "Low Battery" alarm in the status byte of each subsequent frame.



The low battery alarm automatically turns off when the battery is replaced or when the temperature conditions are favorable for the proper functioning of the battery.

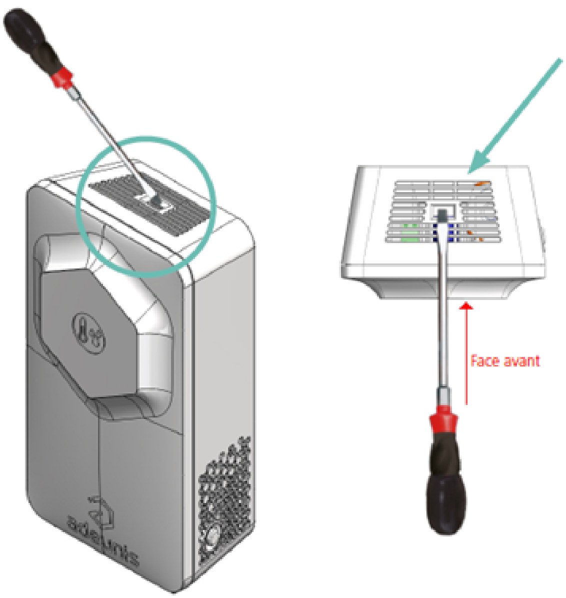
1.8.3 Battery pack replacement

When the low battery indicator is activated, it is possible to change the internal battery pack of the housing.

It is important to use the same battery reference, which is ER18505H.

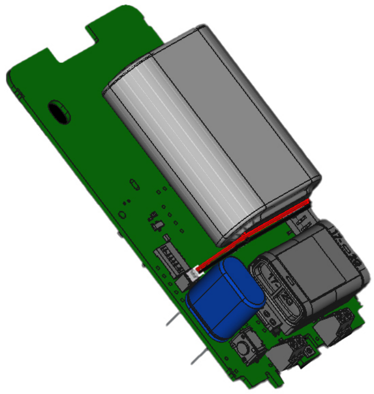
BATTERY PACK REPLACEMENT PROCEDURE:

1. Insert a flathead screwdriver angled towards the front panel into the opening at the top and press down. The housing will open in two parts, and the front panel will detach

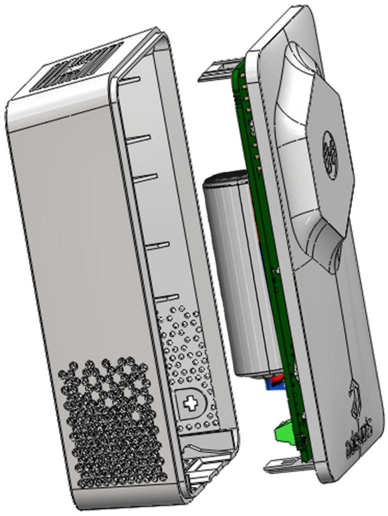


CAUTION: Press gently to avoid damaging the housing

2. Remove the existing battery and replace it with the new one, ensuring that the polarity indicated on the electronic board and the battery model are correctly aligned



3. To close the housing, insert the bottom of the front panel first and snap the top of the housing shut



4. Restart the product

If the product was in PARK mode before changing the battery, press the button. After this procedure, the product will behave as during initial startup. If the product was not in PARK mode during the battery change, it will automatically detect it after sending a few frames and clear the low battery indicator in the status byte

2. TECHNICAL SPECIFICATIONS

2.1 General specifications

Mechanical Specifications	
Dimensions	111 × 61 × 40 mm
Weight	107 g
Housing	IP20 Indoor use
Mounting System	Wall-mounted

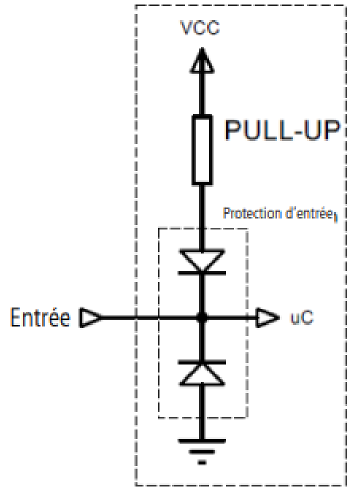
Electrical Specifications	
Supply Voltage	3.6V nominal
Power Supply	Replaceable FANSO Li-SOCl2 Battery Pack - ER18505H
Battery Capacity	4000 mAh

Operating Conditions	
Operating range	0°C / +50°C
Altitude during operation	2000 m or less

RF Specifications	
LoRaWAN Region	EU 863-870 MHz
LoRaWAN Specification	1.0.4
Maximum Transmit Power	+14 dBm
Sensitivity at SF7	-130 dBm
Application Port (downlink)	1
Timestamp Daily Drift between [-10°C and 60°C]	< 3 seconds per day

2.2 Digital Input Interfaces

The block diagram of the digital input interfaces is as follows:



Electrical Specifications	Value
Minimum Absolute Input Voltage	-0.7 Vp
Maximum Absolute Input Voltage	+50 Vp
Recommended Minimum Input Voltage	0 Vp
Recommended Maximum Input Voltage	+24 Vp
Equivalent Input Resistance	500 kΩ
Input Frequency	10 Hz
Current Consumption at HIGH Input Level (>1.5 Vp)	0 µA
Current Consumption at LOW Input Level (<0.7 Vp)	6 µA

Extended operation beyond the min/max values will damage the product.

2.3 Integrated Sensor Specifications

Specifications		Value
Temperature	Technology	CMOSens
	Recommended Operating Range	0°C / +65°C
	Accuracy	± 0.2°C typ. ± 1°C max.
	Typical Resolution	0.1°C
Humidity	Technology	CMOSens
	Recommended Operating Range	10% / 90% RH
	Accuracy	± 2% RH typ. ± 4% RH max.
	Typical Resolution	1% RH

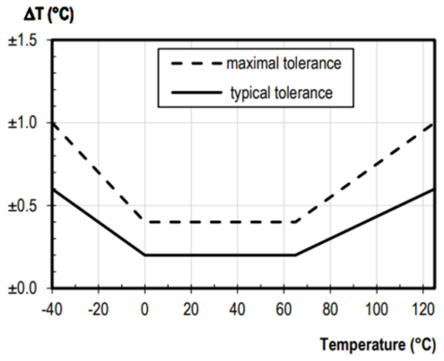


Figure 1 : Précision de la mesure de température

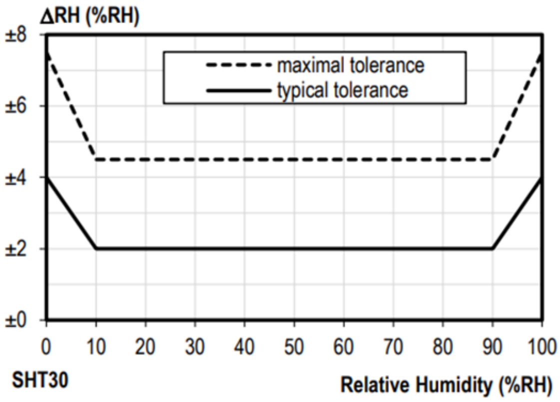


Figure 2 : Tolérance d'humidité relative à 25°C

3. DEVICE AUTONOMY

Usage Conditions:

The table below shows estimated COMFORT autonomy depending on network quality and data transmission frequency.

Measurement and transmission frequency	Number of frames per day	Autonomy SF7 (years)	Autonomy SF12 (years)
Toutes les 10 minutes	144	> 15	2.1
Toutes les 15 minutes	96	> 15	3.1
Toutes les 20 minutes	72	> 15	4.1
Toutes les 30 minutes	48	> 15	5.8
Toutes les heures	24	> 15	10.4
Toutes les 2 heures	12	> 15	> 15
Toutes les 4 heures	6	> 15	> 15
Toutes les 8 heures	3	> 15	> 15
Toutes les 12 heures	2	> 15	> 15
Toutes les 24 heures	1	> 15	> 15

These values are estimations made under certain conditions of use and environment. They do not under any circumstances represent a commitment on the part of Adeunis.

- Storage of the product before use for a maximum of 1 year
- Calculations made at 25°C indoor temperature

- Button deactivated and the digital input deactivated
- The provided estimates do not include data logging. Data logging enables the user to extend the interval between two transmissions, further optimizing the product's autonomy.

📌 Maximum number of samples per frame:

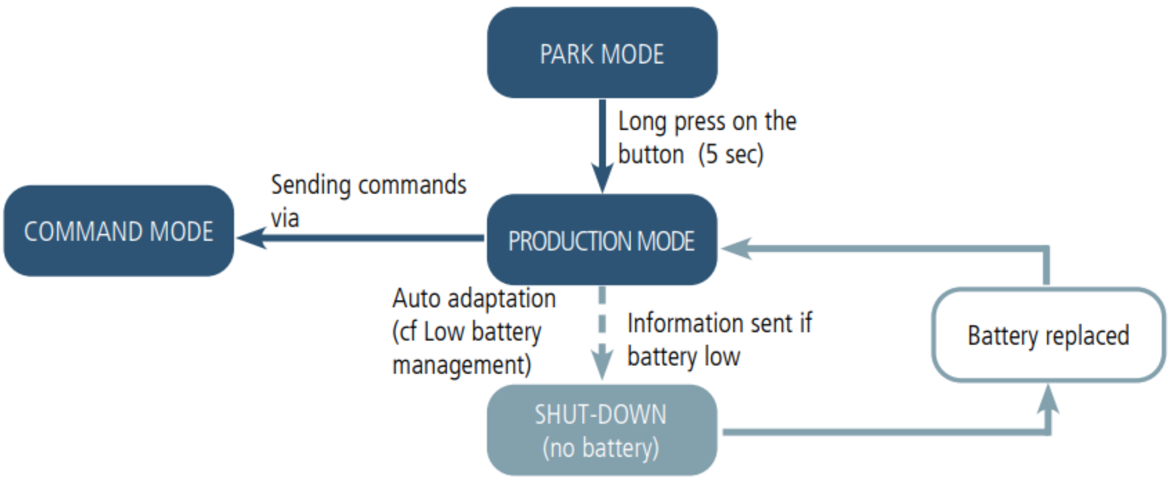
- 16 with timestamping deactivated
- 15 with timestamping activated

4. DEVICE OPERATION

4.1. Operating modes

IMPORTANT NOTE: Adeunis uses the Big-Endian data format

The device has several operating modes :



4.1.1 PARK mode

Le produit est livré en mode PARC, il est alors en veille et sa consommation est minimale. La sortie du mode PARC s'effectue par l'appui sur le bouton latérale du capteur pendant une durée supérieur à 5 secondes. La LED verte s'allume pour signifier la détection de l'appui bouton et clignote ensuite rapidement pendant la phase de démarrage du produit.

Le dispositif effectue sa calibration au démarrage et envoie ensuite ses trames de configuration et de données.

4.1.2 COMMAND mode

This mode allows you to configure the registers of the product.

Please note that it is necessary to install the official Silabs driver available here: **link to Silabs driver**

To enter this mode, connect a cable to the USB-C port of the product and use the IoT Configurator.

To exit COMMAND mode, either use the disconnect function in the IoT Configurator or unplug the USB-C cable.

The product will then return to its previous mode, either PARK or PRODUCTION.

4.1.3 PRODUCTION mode

This mode allows the user to operate the product in its finale use.

4.2. JOIN phase

4.2.1 Start-up of the product, JOIN process and configuration

The product start the JOIN process after entering PRODUCTION mode (after the detection of the magnet or after the exit of the command mode).

By default, the device make 10 successive trials, in case of failure the device waits for 12 hours and then restarts the process. This process will be repeated until the device receive an accept from the gateway called Join Accept.

It is possible to configure the JOIN process through the IoT Configurator. With the App you can decide :

- How many trials you want for each authentication attempt,
- The delay maximum between 2 attempts,
- The weighting factor, used to reduce the delay for the first attempt

ENRegisters concerned by the configuration:

- S312: Maximum delay between 2 authentication attempts
- S313: Weighting factor for initial authentication attempts
- S314: Number of tries for each authentication attempt

  Example:

Register	Encoding	Value	Result
S312	0x2A30	10800	The maximum delay between each attempts is 4 hours.

Register	Encoding	Value	Result
S313	0x04	4	The weighting factor indicated that the first attempt will be spaced by 1 hour, then it will increase after each attempt until it reaches the maximum delay specified in S312.
S314	0x0F	15	Each attempt is composed by 15 successive trials

4.2.2 Launch a JOIN process remotely

The product receives a 0x48 downlink frame and restart after a defined delay (indicated in the frame).

This function of restart enables the device to start a JOIN process remotely. It can be useful for a change of operator or when you have to restart a gateway.

To know the content of the 0x48 frame refers to the Technical Reference Manual (TRM) of the product.

4.3 Network quality test

During the JOIN Process, a device configured in Class A OTAA will make a network quality test (patented algorithm). When the test is running the device shows the 2 LEDs green and red simultaneously (from 10 to 20 seconds).

The result of the test is given by the devices after around 20 seconds following the Join Accept. It is visible through the sole thanks to the LED.

With this information the installer know the quality of the network and can move the product to a place with a better coverage. In any case, the product will send the first frames directly in the SF determined by the result of the test.

4.4. Transmission modes

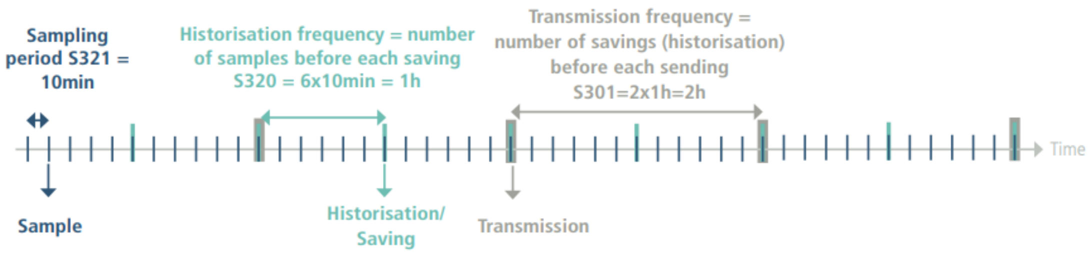
The device can measure the temperature and humidity in a room, save this information and send it in three transmission modes.

	Periodic transmission	Transmission over threshold	Periodic transmission and over threshold
Definition	Periodic sending allows data to be collected in a specified period of time, to be saved and sent on a regular basis for analysis over time.	The sending of a frame on exceeding a threshold makes it possible to read data according to a given period and to send an alarm only if one of the thresholds is exceeded	Mix of the two modes in order to be able to reading regularly to receive alerts if the threshold is exceeded and to save the information regularly to make the analysis over time.
Practical example of use	I want my device to read the tempera- ture and humidity every 3 hours, this in- formation is saved and all my backups are sent to me once a day.	I want my device to send me an alarm when 24°C in my room is exceeded with a reading every 10 minutes. I do not want an alarm for humidity.	I want my device to run every 10 minutes. I want the data saved every 3 hours and the information sent to me once a day. If the temperature exceeds 24°C I want an alarm sent to me. I do not want an alarm for humidity.
Associated configuration	• **Period of acquisition** (S321) = 5400 (5400 seconds = 3 hours) • **Backup frequency** (S320) = 1 (1 backup every 3 hours) • **Frequency of transmission** (S301) = 8 (8 X 3 hours = 24 hours) • Type of alarm T° (S330) = 0 (alarm disabled) • Type of humidity alarm (S340) = 0 (alarm disabled)	• **Period of acquisition** (S321) = 300 (300×2 = 10 minutes) • **Frequency of transmission** (S301) = 0 (no periodic sending) • High threshold definition (S331) = 240 (+24°C) • Type of alarm T° (S330) = 1 (high threshold) • high threshold hysteresis (S330) = 20 (2°C) My room will have returned to the “normal” temperature below 22°C. • Type of humidity alarm (S340) = 0 (alarm disabled)	• **Period of acquisition** (S321) = 300 (300×2 = 10 minutes) • **Backup frequency** (S320) = 18 (18 × 10 min =3 hours) • **Frequency of transmission** (S301) = 8 (8 X 3 hours = 24 hours) • High threshold definition (S331) = 240 (+24°C) • Type of alarm T° (S330) = 1 (high threshold) • high threshold hysteresis (S332) = 20 (2°C) My room will have returned to the “normal” temperature below 22°C. • Type of humidity alarm (S340) = 0 (alarm disabled)

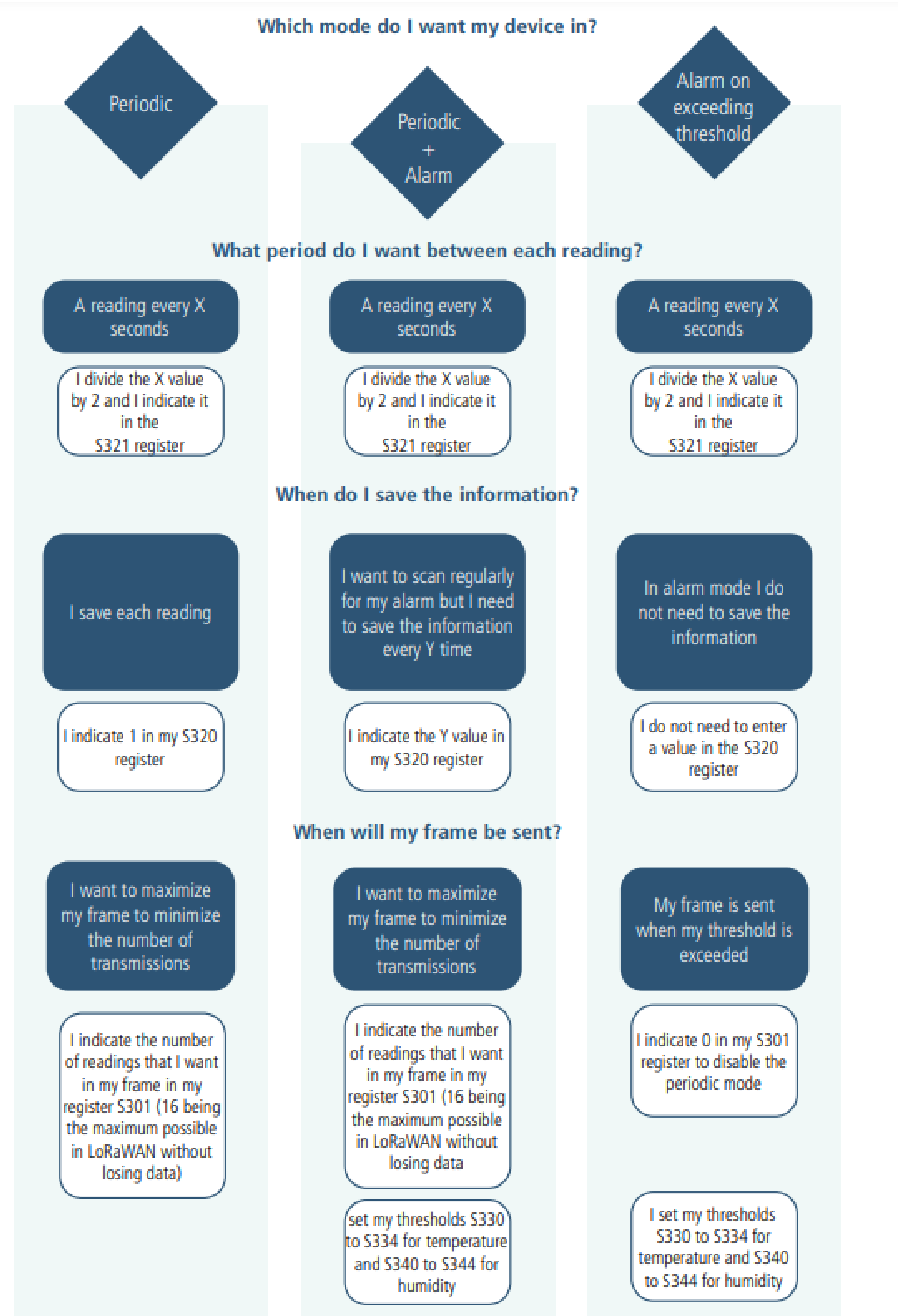
CAUTION: The information sending capacity will depend on the network used. Here the case considered works with a technology LoRaWAN.

PAY ATTENTION: the frame capacity depends of the network used.

Illustration of the mixed mode, with periodical data and sampling for alarm:



Procedure to follow to program its registers according to the chosen mode :



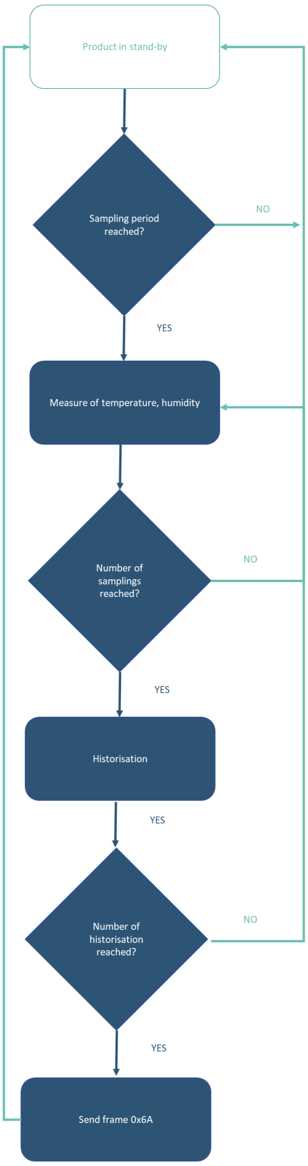
Example of possible configurations:

Desired Case (excluding 100% event-based)	Associated Configuration	Theoretical Number of Periodic Frames Sent per Day
Reading: 10 minutes Backup: every hour (every 6 readings) Sending: every half day (every 12 backups)	S321 = 300 S320 = 6 S301 = 12	2 frames
Reading: 10 minutes Backup: at each reading Sending: maximum tolerated by my frame (here, LoRaWAN)	S321 = 300 S320 = 1 S301 = 15	9 or 10 frames
Reading: 5 minutes Backup: every 15 minutes (every 3 readings) Sending: every 30 minutes (i.e., every 2 backups)	S321 = 150 S320 = 3 S301 = 2	48 frames
Reading: every hour Backup: at each reading Sending: at each backup	S321 = 1800 S320 = 1 S301 = 1	24 frames
Reading: every hour Backup: at each reading Sending: every 4 hours (every 4 backups	S321 = 1800 S320 = 1 S301 = 4	6 frames
Reading: every hour Backup: at each reading Sending: every 10 minutes (every 10 backups)	S321 = 30 S320 = 1 S301 = 10	144 frames

4.4.1 Periodic sending with or without history logs

The device allows the measurement and the periodic sending of the sensor values according to the following diagram:

The device makes it possible to record the temperature and the humidity at a certain frequency, to store this information and then to send it periodically.



The parameters associated with this operating mode are:

- Period of acquisition (S321)
- Backup period (S320)
- Period of transmission(S301)

Example:

Register	Value encoding	Value	Result
S321	Decimal	5400	1 readi 3 hour 2sec = 1080 3 hour
S320	Decimal	1	1 back each r
S301	Decimal	8	Periodi with a 8 ∗ 3 h 24 hou
S330	Decimal	0	Tempe alarm
S340	Decimal	0	Humidi off

In this example:

- The device takes temperature and humidity every 3 hours and saves the information
- The device will make 8 backups and send them once a day
- The device is in pure periodic sending mode since the alarms have been

ADVICE FROM ADEUNIS: By default, the device is set to read every hour (S321 = 1800). For pure periodic sending it is advisable to set the acquisition period to the desired back- up period in order to gain autonomy (here 5400 corresponding to 3 hours).

Be careful about backup and sending values that will also depend on the network used and its bandwidth.

Note: for a transmission without history, it is sufficient to set the register 301 (transmission period) to 1 so the device will send a frame to each backup.

4.4.2 Periodic transmission with redundancy

The redundancy mechanism allows the user to add previous data in the current frame (see here under example). Thanks to this mechanism the device keep a number of samples in its local memory to send them again in the new frame.

The associated parameters linked to this mode are:

- Sampling period (register 321), historisation frequency (register 320) and transmission frequency (register 301)
- The number of samples to be repeated in the next frame (register 323).

When redundancy is activated the frame will contain the number of bytes corresponding to the total of samples, that is S301+S323. At the start-up of the device, the redundant signed bytes are completed by zeros until there is memorized samples.

Example with redundancy:

Register	Encoding value	Value	Result
S321	Decimal	1800	1 sample every hour (1800 × 2 seconds = 60 minutes)
S320	Decimal	1	1 historisation at each sampling
S301	Decimal	1	Transmission of the frame after each historisation
S323	Decimal	1	1 redundant sample per frame

In this example:

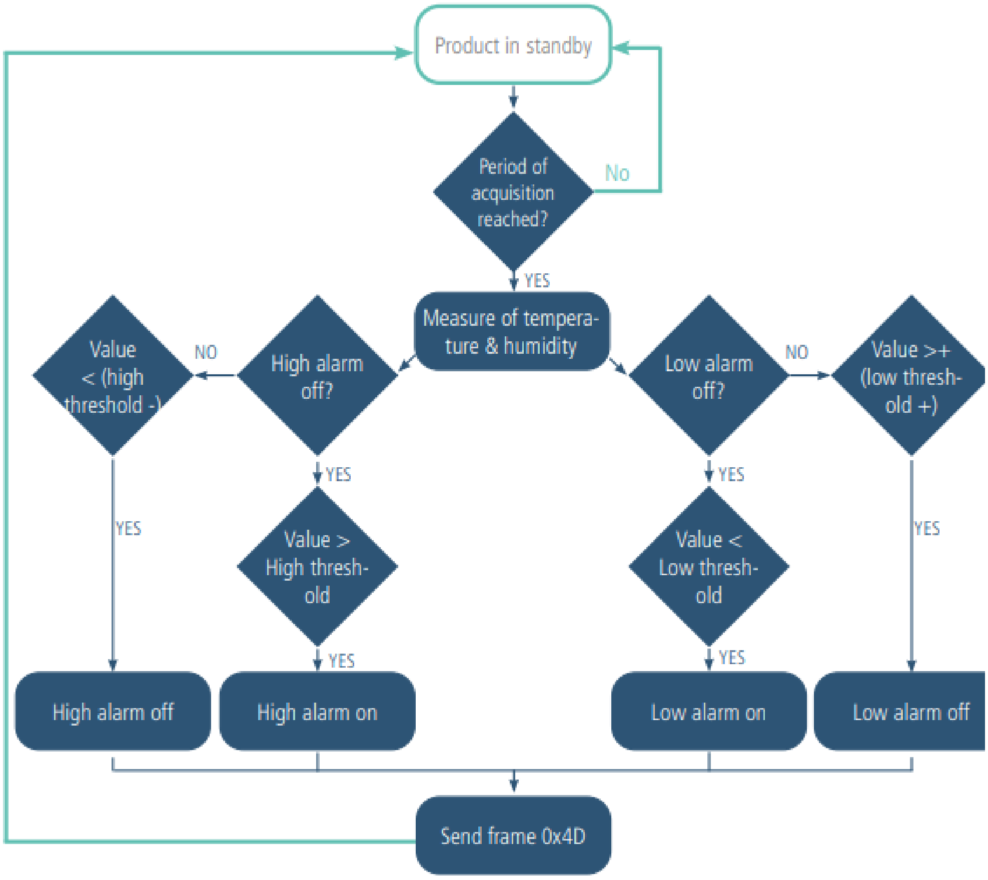
- 1 sampling done every hour (1800×2seconds = 60 minutes)
- The historisation is made at each sampling
- The transmission is made at each historisation so every hour
- The device will send a frame with 1 new sample and 1 memorized sample.

In this example, thanks to redundancy, if a frame is lost, the data is recovered in the next frame.

4.4.3 Sending on exceeding threshold

The device allows detection of exceeding threshold (high and low) for each sensor according to the following schema: The device sends a data frame when a threshold is exceeded but also when returning to normal.

E.g.:



Register	Value encoding	Value	Result
S301	Decimal	0	Event mode (no periodicity)
S321	Decimal	300	One reading every 10 minutes (300/60 sec x 2)
S330	Decimal	2	Alarm type for high threshold temperature
S331	Decimal	240	Temperature at + 24°C (240/10)
S332	Decimal	20	Hysteresis at 2°C (20/10) below the high threshold of 22°C
S340	Decimal	0	Humidity alarm off

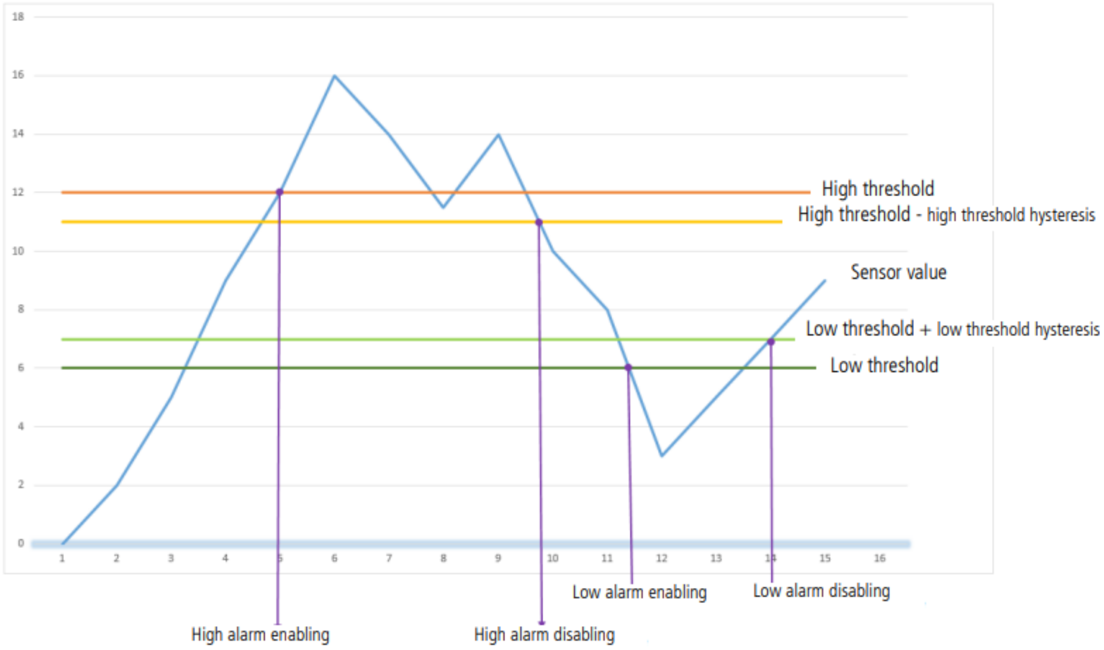
In this example:

- The device takes temperature and humidity every 10 minutes
- The device will trigger an alarm if the temperature is above 24°C, no alarm indicated for humidity

• The alarm will be disabled if the temperature drops below 22°C

NOTE: As described, it is possible to combine the periodic mode and the alarm mode.

Explanation of thresholds and hysteresis



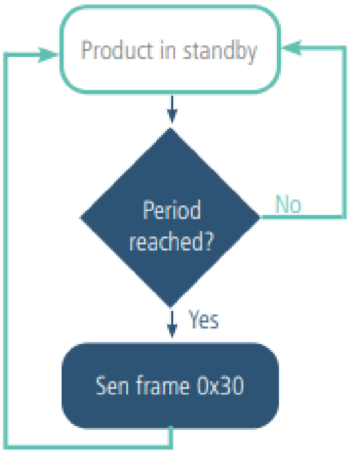
The parameters associated with this operating mode are:

- The transmission period (equal to zero in this use case) (register 301).
- The acquisition period (register 321).
- The high alarm threshold for the temperature sensor (register 331).
- The high alarm hysteresis for the temperature sensor (register 332)
- The low alarm threshold for the temperature sensor (register 333).
- The low alarm hysteresis for the temperature sensor (register 334).
- The high alarm threshold for the humidity sensor (register 341).
- The high alarm hysteresis for the humidity sensor (register 342).
- The low alarm threshold for the humidity sensor (register 343).
- The low alarm threshold for the humidity sensor (register 344).

4.5 Transmitting the Keep Alive frame

If the device does not have periodic data configured, and no threshold is exceeded, it may not transmit data for a long time. So, to be sure that the device is working properly, it transmits a Keep Alive frame (0x30) according to a determined frequency (S300)

The parameters associated with this operating mode is the setting of the transmission period of the Keep Alive frame (register 300).

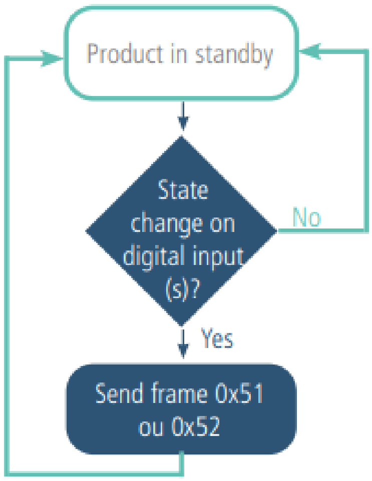


E.g.: I want a Keep Alive frame sent to me every 24 hours

Register	Value encoding	Value	Result
S301	Decimal	0	Disabling periodic sending
S300	Decimal	8640	8640×10 = 86 400 seconds 1440 minutes or 24 hours

4.6 Alarm on the digital input

The device incorporates two digital inputs, one through the connected button and one via the terminal block, both for detecting a change in up and down state.
The device allows the sending of a frame following a change of state on one of its inputs according to the following diagram:



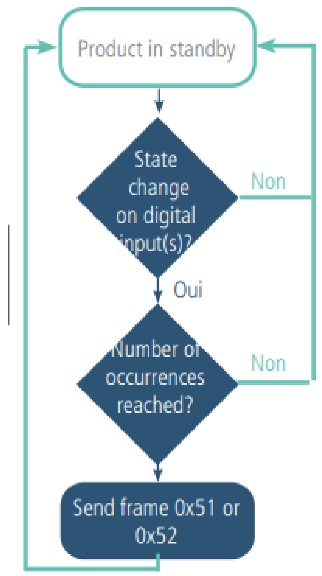
Example :

Register	Value encod- ing	Value	Result
S380	Hexadecimal	0x41	Configuration of the Digital Input 1 (button): • Detection of high edges • Debounce time* 100ms
S381	Decimal	1	The device sends a frame every time the button is enabled
S382	Hexadecimal	0x0	Configuration of the Digital Input 2 input (terminal block): • Disabled • No debounce time

*Debounce time: minimum time to take account of a change of state. For example, if this period is 10 ms all pulses (high or low level) whose duration is less than 10 ms will not be considered. This technique avoids potential rebounds during a change of state.

In this example the device:
• The device has a debounce time of 100ms and the button press alarm is enabled (register 380).
• The device sends a frame for each button press (register 381)
• The alarm via the terminal block is disabled (register 382)

NOTE: It is possible to program the sending of a frame only after a certain number of edge detections (S381/S383).



Register	Value code	Value	Result
S382	Hexadecimal	0x41	Configuration of the Digital Input 2 input (terminal block): • Detection of high edges • Debounce time* 100ms
S383	Decimal	5	The device sends a frame every 5 times that a high edge is detected on Digital Input 2

*Debounce time: minimum time to take account of a change of state. For example, if this period is 10 ms all pulses (high or low level) whose duration is less than 10 ms will not be considered. This technique avoids potential rebounds during a change of state.

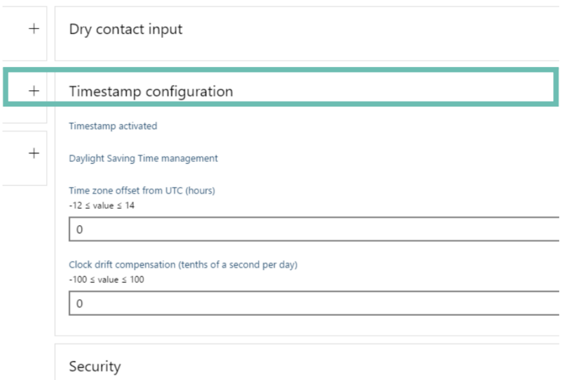
In this example the device:
• The device has a debounce time of 100 ms and the button press alarm is enabled (register 383).
• The device sends a frame as soon as it has detected 5 high edges on its digital input per terminal block (register S382).
The digital input operates only in event mode (no periodic sending).

4.7 Timestamp of the data

The sensor can integrate the timestamp of the data in the frame if this option is activated. Timestamp is given in EPOCH 2013 (please, refer to the TRM of the product to know the content of each frame).

To configure the timestamp, you have to set the UTC time first, via Downlink or through the Advanced Menu of the IoT Configurator.

Then you an activate the timestamp in the Applicative parameters and choose if you want to set the timezone and if you want that the product take into account the Daylight Saving Time.



5. REGISTERS AND FRAMES

To know the content of the registers and of each frames (uplink and downlink) of the product, refers to the TECHNICAL REFERENCE MANUAL of the product, available on the adeunis website: https://www.adeunis.com/en/produit/comfort-temperature-humidity-2/

6. CONFIGURATION AND INSTALLATION

6.1 Transmitter configuration and installation

To configure the product, it is recommended to use the IoT Configurator (android and Windows application).

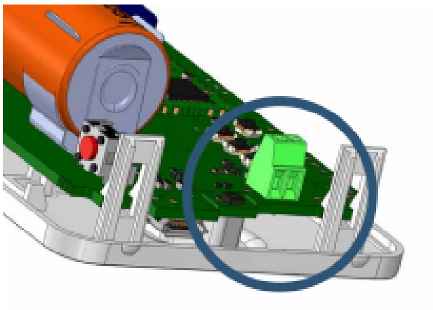
- Google Play: https://play.google.com/store/apps/details?id=com.adeunis.IoTConfiguratorApp
- Windows 10: https://www.adeunis.com/telechargements/

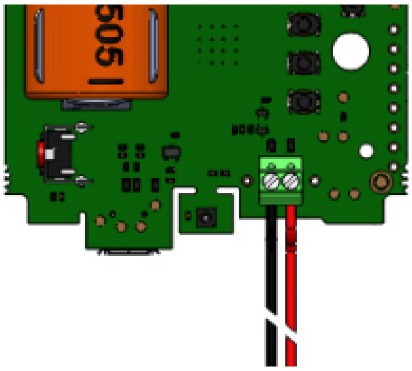
The product can also be configured remotely via the network by sending downlink frames to it. For this, refer to the product's TECHNICAL REFERENCE MANUAL, available online on the product page:

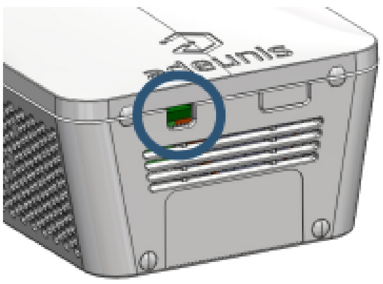
https://www.adeunis.com/produit/comfort-temperature-humidite-2/

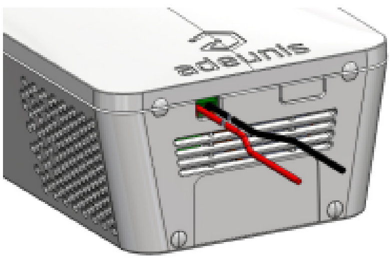
To configure the product using AT Command or radio recommendations, please refers to the INSTALLATION GUIDE Adeunis.

6.2 Wiring the Digital Input via the terminal block









In order to be able to combine a dry contact sensor with the device and thus benefit from its Digital Input, it is necessary to connect the sensor to the terminal block of the card.

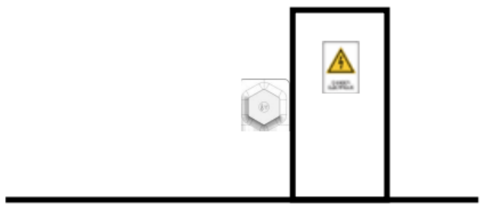
Connection procedure for the wires:

1. Open the unit
2. Plug the two wires into each notch of the terminal block
3. Break the element of the casing so as to pass the wire onto the back of the casing
4. Configure the digital input alarm
5. Close the unit
6. Restart the device with the button as for a first start

After this procedure the device will behave as during a first start

E.g.:

The LoRaWAN COMFORT device can easily be combined with a cabled door contact via the terminal block (Digital Input 2). Thus, positioned next to the door of a secure room under control, the device will be able to send an alarm every time the door is opened/closed and thus enable the security manager or the building manager to verify compliance with the security requirements on the site.



6.3. Product Installation Recommendations

Recommended positioning for use in tertiary and domestic environments:

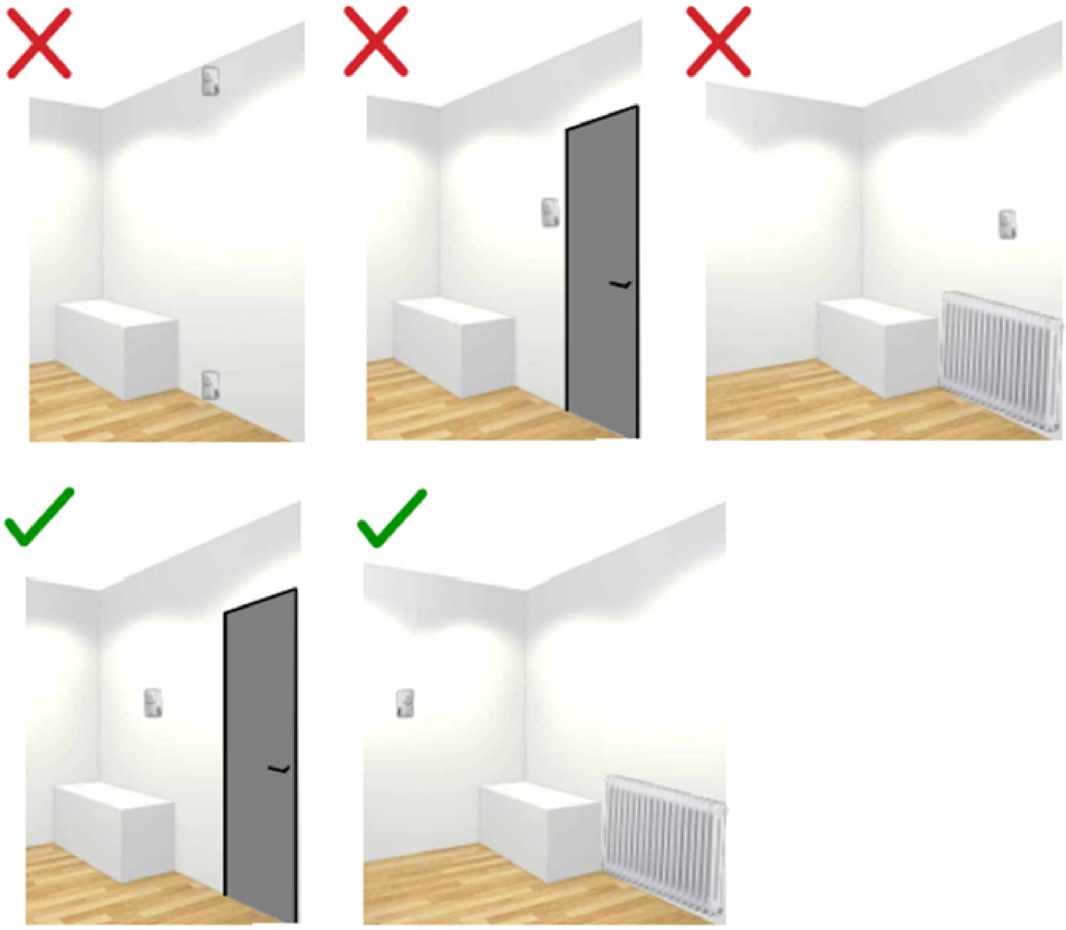
- Mounted on a wall at a height of 1.5 meters from the ground, in an unobstructed area and not exceeding 2 meters.

The product should be installed using 2 screws CBLZ 2.2 × 19mm and 2 plugs SX4. Use these products or equivalent ones to secure your product to a flat support.

To obtain accurate air quality measurements, we recommend following the instructions below:

- Do not use the sensor for outdoor purposes.
- Do not position the sensor facing the sun, near a heat source, cold source, ventilation, or in a draft.
- Maintain a minimum distance of one meter from a door or window.
- Do not install the sensor too close to the ground, below 30 cm.
- Do not install the sensor in a dusty or poorly maintained area (garage, basement, workshop, etc.).
- Do not install the sensor in areas with consistently high humidity levels exceeding 95% relative humidity (bathroom, changing rooms, spa, laundry room, etc.).
- Do not install the sensor in an area where it can be damaged or torn off.

ATTENTION
The top face of the product (allowing the opening of the housing) must be accessible with a screwdriver. Do not position it against a ceiling or under an object that may prevent opening the housing.



7. DOCUMENT HISTORY

Version	Content
V1.0	Creation